Korea Technology and Information Promotion Agency for SMEs

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Innovative, non-destructive testing and evaluation of joints, such as thermal connection of battery cells or adhesive bonds**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery (√), Advanced Mobility (√), Next-gen Nuclear Power (√), Advanced Bio ( ), Aerospace/Marine (√), Hydrogen (√), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical and materials		*Electrical and Electronics*	
	Medium Category	Small Category	Medium Category	Small Category
	Batteries	Physical Quantity Test/Analysis Instruments	Batteries	Manufacturing and Measurement/Evaluation equipment
Concept of Proposed Technology	Joints between two joining partners, such as thermal connections between battery cells or adhesive bonds, play a decisive role in the quality and service life of technical systems. This makes reliable production processes essential, and these processes must be properly monitored. The same applies to the inspection of joints after repair work, for example when individual cells in a battery are replaced. A new non-destructive measurement and testing method for joints has been developed at laboratory scale at the ZFW. The method is based on the thermal transient method, which has been successfully applied in power electronics in recent years [1]. In this method, the joint is subjected to a thermal excitation from one side. For this purpose, an inspection stamp is applied to the accessible side of			

the joint. In the case of a battery, this could be the outer surface of the battery housing. The inspection stamp is heated using a step function, while the resulting temperature change at the stamp is measured simultaneously. Fig. 1 shows an example of the step heating of a thermal connection of a battery cell.

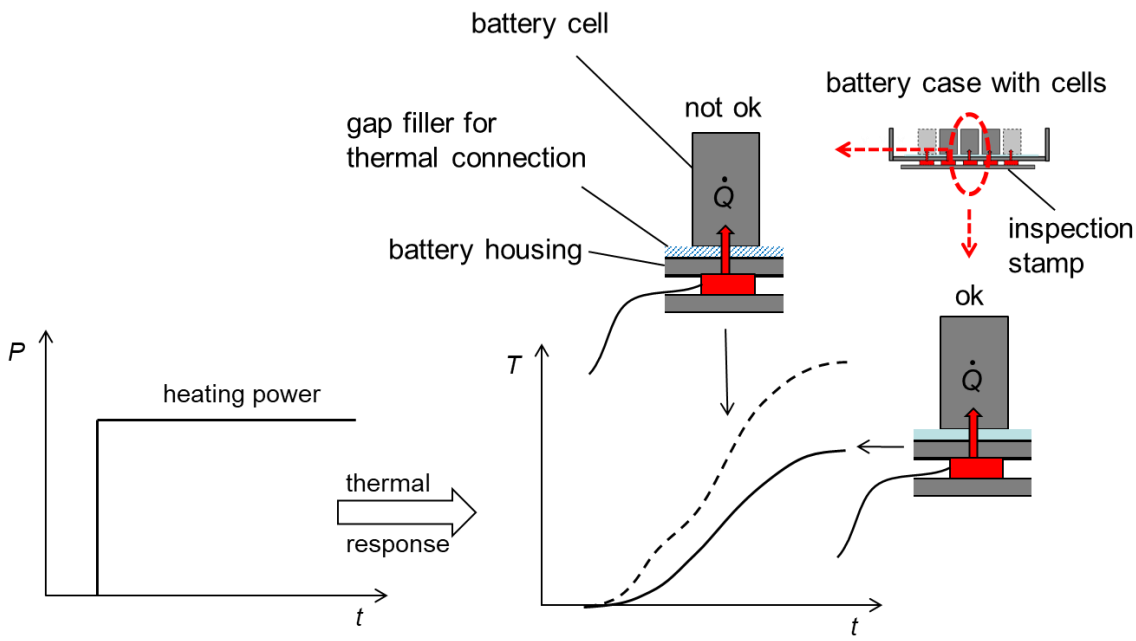


Fig. 1: Heating of the thermal connection of a battery cell with a step function and measuring of the thermal response.

The thermal response *T(t)* depends on the condition of the thermal interface material (gap filler). If the cell is well connected to the housing, the temperature rise is smaller (solid line, Fig. 1) than in the case of a non-ideal connection (dashed line). The measurement not only allows a simple “good–bad” assessment, but also provides a quantitative evaluation of the joint quality.

For this purpose, the thermal response *T(t)* is transformed into a so-called structural function using a dedicated mathematical algorithm [1]. This yields the thermal resistance R_{th} of the connection in K/W. Deviations between the measured and expected R_{th} values can have various causes, including process errors during assembly, air bubbles or defects in the gap filler, and insufficient adhesion of the gap filler to the surfaces due to contamination.

Changes in the thermal resistance R_{th} caused by aging effects can also be quantified. Possible mechanisms include the well-known pump-out effect, as well as material embrittlement or degradation.

The advantages of the new measurement method over other methods are:

- The method is non-destructive.
- It is based on thermal analysis and therefore does not rely on the electrical conductivity of the test specimens.

	• It provides a quantitative value of the condition of the joint connection, not only qualitative results. This is completely new and enables a precise assessment of the joint. • For the measurement, one-sided contact with the structure is sufficient. This is a particular advantage when the opposite side is inaccessible, such as with battery housings and battery cells. • Depending on the structure of the samples, the measurement takes only a few seconds. • The measurement method can be easily used for process control in manufacturing. The functionality of the method has been demonstrated in principle under laboratory conditions. The proposed project aims to further develop the method for practical industrial application. The measurement method has already been patented in Germany [2].
Rationale for Joint R&D Project	The new measurement method offers significant potential for the non-destructive testing of joints. It provides fast and reliable information on the condition of a connection during manufacturing or quality assurance. The functionality of the method has been demonstrated in principle; however, substantial further development is required for practical implementation. This includes adapting the test stamp to real application conditions (e.g. inspection stamp geometry suitable for the sample geometry and thermal conductivity of the joining partners), further development of the mathematical evaluation algorithm, and ultimately the design and construction of a compact, market-ready testing device. The project therefore requires two partners. ZFW as research institute will be responsible for the theoretical aspects, such as the mathematical framework, while the Korean industrial partner will cover the practical side, including real-world application requirements (see Chapter 4, Roles and Responsibilities, for details). Only the close integration of theory and practice can ensure the success of this project and enable the development of an innovative, complex, and robust measurement method for industrial use.
Key Words	1. Non-destructive testing 2. Testing of gap fillers in batteries 3. Non-destructive testing of adhesive bonds 4. Thermal testing in production 5. Quality control.

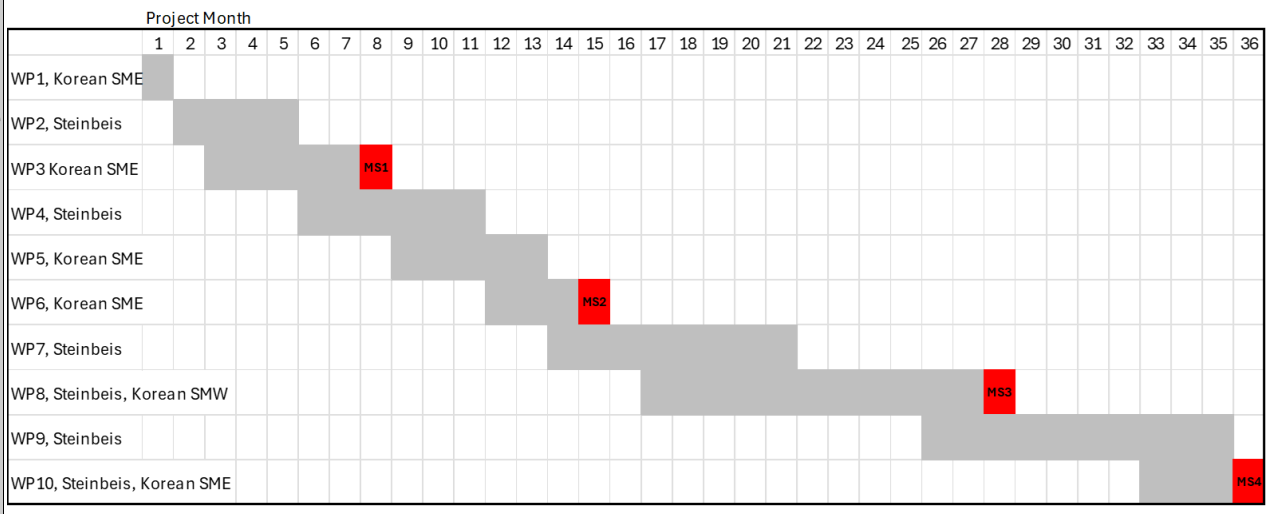
3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level 3	End of the Project	Level 9
R&D Objectives	Based on a laboratory setup and a fundamental evaluation mechanism, a new, highly innovative measuring and testing device is to be developed.			
R&D Target Specifications	The thermal resistance (R_{th}-value) is used as a key parameter for evaluating the condition of a joint. It characterizes the heat transfer across the contact interface: the poorer the connection, the higher the thermal resistance. One of the main features of the new measurement system is its high sensitivity. Even very small deviations from an ideal reference joint should be detectable. For this purpose, a thermal resistance resolution of R_{th} = 0.01 KW is targeted. In addition to the main device, a crucial component is the inspection stamp. It must be precisely adapted to the sample under test and should enable simple, potentially even automated, handling. The measurement result must be independent of the mechanical contact conditions between the inspection stamp and the sample surface. In particular, the contact pressure of the stamp must not influence the measured thermal resistance at the interface. This is ensured partly by the mathematical evaluation algorithm and partly by a stamp design that allows for defined and reproducible contact conditions.			

R&D Plan	Work Package (WP) 1:	Definition of the specimens with the joint to be evaluated Definition of good and bad parts with the corresponding deviation in thermal resistance, definition of necessary clock rate and other special measurement requirements (e.g. surface aspects, aging effects).
	Work Package (WP) 2:	Development of an initial inspection stamp Based on the information from WP1, the first variant of an inspection stamp is developed. For this: Determination of the necessary measurement parameters as heating power, measurement time, required temporal resolution, requirements for voltage measurement, performing thermal simulations (software FloEFD or ANSYS) for the dimensioning of the stamp.

	Work Package (WP) 3:	Construction of the initial stamp Procurement of materials and construction of the initial stamp.
	Work Package (WP) 4:	Adaptation of the measurement and evaluation algorithm Adaption of the measurement algorithm to the samples to be measured (result of WP1), programming of an initial control and measurement software for measurement and data acquisition using LabView (key points: dynamic of power supply and data acquisition adapted to sample, signal-to-noise ratio of measured step-response curve *T(t)*.
	Work Package (WP) 5:	Preparation of test samples Construction of test specimens with joints of varying quality, this includes ideal samples, defined poor samples, and targeted gradations in between.
	Work Package (WP) 6:	Conducting the first test measurements Initial test measurements with the new inspection stamp and the new software. Test of measurement sensitivity and reproducibility.
	Work Package (WP) 7:	Construction of the base unit with electronics, control unit, and housing Design and construction of the microcontroller-based measuring device (basic unit), including the interfaces for the computer and inspection, development of mechanical and electronic components, assembly and basic tests.
	Work Package (WP) 8:	Development of the second generation of inspection stamps Based on the experience gained in WP1-8, an improved version of the inspection stamp will be developed and constructed. The new development lounge will be extensively tested with samples from WP5.
	Work Package (WP) 9:	Fine-tuning of hardware and software In a further round, final optimizations will be made to the hardware and software. To this end, further measurements will be carried out on laboratory samples. For the first time, the measurement method will also be tested in practical operation by the Korean company. Optimizations will be made where necessary.
	Work Package (WP) 10:	Documentation of the project The project will be documented in detail. This includes design drawings and circuit diagrams, as well as an operating manual.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The Steinbeis Transfer Centre for Heat Management in Electronics (ZFW) is primarily focused on the theoretical aspects of the project. This mainly includes WP2 (Development of an initial inspection stamp), WP4 (Adaptation of the measurement and evaluation algorithm), WP7 (Construction of the base unit including electronics, control unit, and housing), WP8 (Development of the second generation of inspection stamps), and WP9 (Fine-tuning of hardware and software).
Expected R&R of Korean SME Partner	The Korean SME partner is responsible for the more application-oriented work packages. This includes WP1 (Definition of the specimens with the joints to be evaluated), WP3 (Construction of the initial inspection stamp), WP5 (Preparation of test samples), WP6 (Conducting the initial test measurements), and WP8, (Development of the second generation of inspection stamps), which is done together with Steinbeis. Both partners share responsibility for WP10 (Project documentation).
General Comment	The project requires close collaboration between the project partners. To this end, the work packages are tightly interlinked. This enables both partners to benefit from each other's expertise and, ultimately, to jointly develop an innovative measurement method. As a result, the project will significantly enhance the competitiveness of the Korean SME partner on the international market.
Temporal Sequence	The following diagram provides an overview of the project workflow with the following Milestones (MS): MS1: The first stamp is constructed and ready for use MS2: Initial test measurements successfully completed MS3: Second generation of inspection stamp developed and manufactured MS4: The project has been successfully completed and is well documented. 

5. Expected Outcomes/Impact

Expected Outcomes and Impact	At the end of the project, a new, practical measurement method for the non-destructive testing of joints will be available. The method has been developed in the form of a compact, industrial-grade, marketable device. The scope of application is diverse in principle, ranging from production monitoring to quality control. Depending on the application, only the test adapter needs to be adapted. The basic device can remain unchanged. This allows for better control of joining technologies in many industrial areas, such as adhesive bonding processes or thermal bonding with gap fillers. Overall, an enormous economic and business impact is to be expected.

6. References

[1] JEDEC Solid State Technology Association: JESD51-14, Transient Dual Interface Test Method for the Measurement of the Thermal Resistance Junction-to-Case of Semiconductor Devices with Heat Flow Through a Single Path, JEDEC, Arlington, 2010

[2] A. Griesinger, “Method for detecting the state of a connection between components,” German Patent DE102016207527A1, 2016.